

PROGRAM 9

Organize kubernetes pods using labels and namespaces

Project structure

1.namespaces.yaml

2.pods.yaml

Creation of these files

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Dockerfiles/program-9  ×  1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/D
---
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ cat namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod

1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ cat pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Apply the namespaces and pod yaml

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Docke... x 1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Docke... x 1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Docke... x
name: backend-pod
namespace: prod
labels:
  app: backend
  env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80

1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl get pods -n dev
kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          11s
frontend-pod  0/1     ContainerCreating   0          11s
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          11s
frontend-pod  0/1     ContainerCreating   0          11s
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl get pods -n dev
kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          92s
frontend-pod  1/1     Running   0          92s
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          92s
frontend-pod  1/1     Running   0          92s
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0          107s
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          113s
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$
```

Filter pods by label

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl get pods -n dev
kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          92s
frontend-pod  1/1     Running   0          92s
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          92s
frontend-pod  1/1     Running   0          92s
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0          107s
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0          113s
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-9$
```